

Chain of Thought Reduces Errors from Misleading Names in Arithmetic

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Abstract

Asking language models to write out arithmetic calculations reduces answers that follow a misleading variable name. Building on Kudo et al.’s arithmetic study, we test thirteen models on pairs of problems with the same equations. In one version, a variable has an ordinary name; in the other, its name is a word for an incorrect digit. When asked only for the final answer, models more often return the digit suggested by the misleading name. Chain-of-thought prompting asks them to write calculations before answering and reduces this tendency. The reduction is reliable in 10 comparisons; with calculations, 43 of 44 added-error intervals lie within 2 percentage points of zero. Writing the calculation therefore makes answers more robust to misleading wording while the underlying math remains unchanged.

1 Introduction

Consider the problem

pen=1 + cup, cup=2 + 3; pen=?

pen has value 6. Renaming it four leaves the answer unchanged:

four=1 + cup, cup=2 + 3; four=?

Only the label changes: cup is still 5 and the queried variable is still 6. Answering 4 instead follows the name. We call 4 the *name-suggested value* and that answer a *name error*. Because 4 occurs nowhere in the equations and differs from every variable’s value, this error can be distinguished from copying a digit.

Can asking the model to calculate before answering reduce this error? In our *answer-only* condition, the model is prompted to return just the final digit. In our *chain-of-thought* (CoT) condition, it is prompted to restate and evaluate the equations before giving that digit. Here the calculation includes $\text{cup}=2+3=5$ and $\text{four}=1+5=6$. The comparison concerns what the model generates; an answer-

only response can still involve internal computation.

Misleading names already reduce accuracy on code tasks (Le et al., 2025; Lam et al., 2025; Le et al., 2026). We use arithmetic to make the competing answers explicit, building on Kudo et al. (2026)’s study of computation during CoT. We add a name that suggests a different answer while preserving the equations. This conflict, like those in Stroop and shape–texture studies (Stroop, 1935; Geirhos et al., 2019), separates the name’s suggestion from the computed value.

A separate analysis asks which values are readable inside the model. We train a *linear probe*, a small predictor of a variable’s true digit from internal activations, on separate problems. A probe predicting 6 and the model answering 6 are separate observations. We show reliable reductions in name-suggested answers and small remaining effects in almost every tested contrast. The paired renaming comparison makes resistance to a competing word a measurable benefit of CoT while holding the equations fixed.

2 Study design

We use Kudo et al.’s single-digit addition/subtraction task. The main level requires two operations and presents the queried variable’s equation first, before the equation needed to evaluate it. Two models cover all five levels (Appendix A). Matched versions change one name (Table 2): congruent names denote the true value; incongruent names denote another digit, absent from all inputs and variable values. We rename queried and intermediate variables separately. All names occupy one token, preserving matched token positions.

Both prompting conditions use three fixed sets of three worked examples. We evaluate thirteen Llama, Gemma and OLMo checkpoints (Appen-

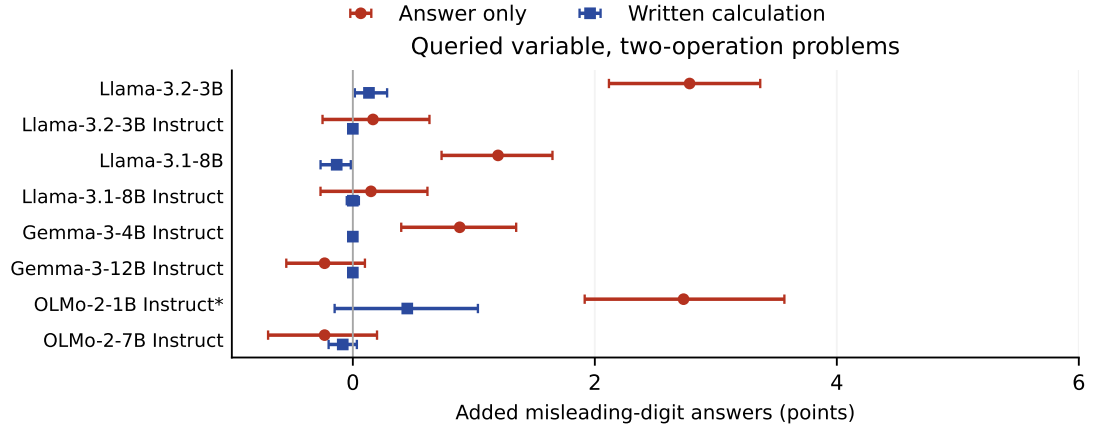


Figure 1: Added answers matching the misleading name, relative to ordinary-name twins. Bars: 95% intervals over three demonstration sets. *Below the planned 90% ordinary-name accuracy floor with written calculations.

dices A, B.3). The main table applies the preregistered 90% mean neutral-CoT accuracy floor; excluded results appear in Table 4.

Excess name errors subtracts how often the ordinary-name version answers the suggested digit from how often the misleading-name version does. This controls guesses when accuracy is low. *Facilitation* is congruent-minus-neutral accuracy; *interference* is neutral-minus-incongruent accuracy. Reliable effects have consistent signs across all three demonstration sets and 95% intervals excluding zero. Equivalence requires a 90% interval inside ± 2 points. Cluster bootstraps resample matched problems with their demonstration repeats (Appendix B.1).

Probes train on separate neutral problems with three seeds. Random digits assigned to names control for learning name identity (Hewitt and Liang, 2019). Patching replaces name activations with a matched twin’s (Zhang and Nanda, 2024); neutral-word and alternative-name controls test disruption and name specificity. Most CoT measurements supply the correct calculation token by token. Generated-chain probes select layers on computation-disjoint validation problems before held-out testing (Appendix F).

3 Results

3.1 Writing calculations limits misleading answers

Figure 1 shows added misleading-digit answers; Table 1 also reports changes in overall accuracy. Without CoT, a congruent name raises Llama-3.1-8B’s accuracy by +4.8 points, and a misleading

name produces +1.2 points of excess name errors on the queried variable. Across levels, the largest effects occur where direct-answer accuracy is near chance; the matched baseline prevents those arbitrary digit answers from being attributed to the name.

With CoT, 43 of 44 name-error contrasts meet the equivalence bound. Paired reductions meet our reliability rule in 10 contrasts; 31 answer-only contrasts already meet the bound (Appendix G). The exception is OLMo-2-1B-Instruct: its intermediate-variable excess is +3.2 points [+2.5, +3.9]. Its neutral CoT accuracy is 47%, compared with 77% under letter names. CoT limits answers matching the misleading digit. Overall accuracy still depends on names: 15 accuracy contrasts remain reliable (Appendix B.1).

3.2 Correct written values can precede name errors

A generated calculation can state the correct value and still end with the suggested digit. Among matched versions that both state the intermediate variable’s correct value, OLMo-2-1B still has +3.6 points of excess name errors [+2.6, +4.7] across 2739 matched observations from 3 demonstration sets (Appendix B.2). This conditional comparison shows that stating the value does not guarantee a correct answer; it does not estimate the causal effect of stating it.

Nor do name errors always repeat the preceding digit. Across 4110 generated CoT name errors, pooled over models, levels and demonstration sets, only 13% have the name-suggested digit at the end of the calculation segment before the final answer

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
Gemma-3-12B-it	CoT	100 [100,100]	+0.0	+0.0	+0.0	+0.0
Gemma-3-12B-it	direct	78 [76,81]	-3.2*	-0.2	-3.7*	+0.4
Gemma-3-4B-it	CoT	100 [100,100]	-0.0	+0.0	-0.0	+0.0
Gemma-3-4B-it	direct	55 [48,59]	+0.2	+0.9*	+0.6	+0.3
Llama-3.1-8B-Instruct	CoT	100 [99,100]	+0.2	+0.0	+0.3	+0.0
Llama-3.1-8B-Instruct	direct	52 [45,65]	+1.7	+0.1	+5.2*	+0.2
Llama-3.1-8B	CoT	95 [86,100]	-1.3	-0.1	+3.8	-0.0
Llama-3.1-8B	direct	53 [41,62]	+4.8*	+1.2*	+3.8*	-0.0
Llama-3.2-3B-Instruct	CoT	100 [100,100]	-0.0	+0.0	-0.2	+0.0
Llama-3.2-3B-Instruct	direct	44 [40,50]	-1.8*	+0.2	+0.5	+0.5
Llama-3.2-3B	CoT	98 [97,100]	-1.2	+0.1	+0.9	-0.2*
Llama-3.2-3B	direct	22 [13,31]	+5.4*	+2.8*	+1.1	+1.8
OLMo-2-7B-Instruct	CoT	95 [93,97]	+0.7*	-0.1	-2.7	-0.0
OLMo-2-7B-Instruct	direct	54 [50,58]	+0.6	-0.2	+3.8*	-0.4

Table 1: Level 3 effects in percentage points: means across three demonstration sets [ranges]. *Helps*: congruent-name accuracy gain; *Errors*: excess name errors. *: consistent signs and 95% intervals excluding zero. CoT name-error contrasts meet the ± 2 -point equivalence bound.

(Appendix B.2). A name error need not copy a trailing instance of that digit.

We also test writing values alone, such as cup=5, four=6, instead of restating and evaluating equations. On two-operation problems, ordinary-name accuracy falls to 54% for Llama-3.1-8B and 31% for Llama-3.2-3B (Appendix B.4). This format changes both arithmetic performance and the written context. It cannot isolate an effect of name repetition while preserving computation.

3.3 What the probes recover

With the correct calculation supplied, Llama-3.2-3B’s probe accuracy just after the intermediate equation is 49.0%, 31.8 points above the name-identity control. At the query, the control reaches 96.9%, indicating name identity. Before the value is written, correct-digit accuracy reaches 97.9%; name-suggested predictions fall to zero.

Across 9 models, 17 of 18 variable readouts have neutral accuracy of 97.6–100.0% and excess name errors within 0.4 points of zero. OLMo-2-1B’s intermediate readout scores 21.8%. The model-mean correlation between misleading-name probe accuracy and excess name errors re-

verses from $r = -0.997$ to $+0.776$ when OLMo-2-1B is omitted (Appendix D). Overall task competence remains a competing account. Within-problem fits also disagree on the association’s sign (Appendix C).

Probes trained on correct calculations score 19.4–62.2% on OLMo-2-1B’s neutral generated chains. Training on generated neutral chains raises this to 50.3–80.1%, still below the 90% calibration minimum in every cell (Appendix Table 15). Llama-3.2-3B reaches 96.3–100.0% with the same design, but its name-error cases remain rare. OLMo’s generated-chain readouts therefore remain descriptive.

3.4 Replacing name representations changes answers

For Llama-3.2-3B without CoT, replacing internal states at misleading names with those from the ordinary-name problem changes 52% of name-error answers to another digit (all layers; Figure 2). Removal peaks at 69% at the defining equation, layers 0–7 (Appendix C). The neutral-word control also turns 22% of correct answers into errors.

Llama-3.1-8B’s neutral and alternative-name patches remove errors similarly. Under sup-

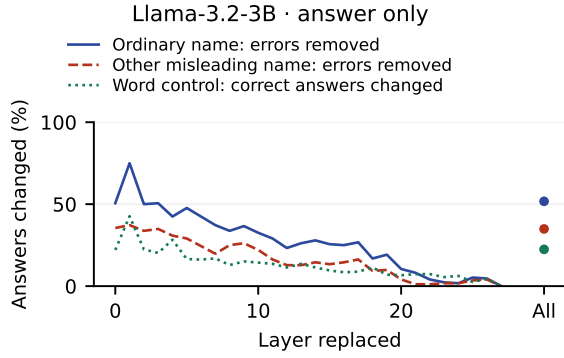


Figure 2: Llama-3.2-3B, queried variable, demonstration set 7. Removal: name-error answers changed to any other digit. Damage: correct answers changed. All: every layer replaced.

plied correct calculations, injecting the misleading name into the exception model’s neutral run moves 2.7% of answers to the suggested digit; the word control changes 0.4% at that layer. These selected interventions do not establish a common causal mechanism.

4 Implications and next experiments

Writing calculations improves resistance to a word that suggests the wrong answer while the mathematical problem stays fixed. This gives reasoning evaluations a concrete test beyond aggregate accuracy: rename a variable and measure whether answers move toward the implied digit. The remaining accuracy differences show why both outcomes matter. The written-value control also distinguishes a correct value in the calculation from a correct final answer: seeing that value alone can miss the effect of a misleading name.

To distinguish calculation from repeated wording, vary whether responses repeat names and include equations, matching length and ordinary-name accuracy. Next, hold a correct generated calculation fixed and vary the final query’s name and digit-only versus named output. The values-only result motivates the accuracy control; the written-value result motivates the final-query test. These comparisons are future work.

5 Related work

Kudo et al. (2026) locate value readouts during arithmetic CoT; our names introduce a competing digit. Language-model Stroop studies test lexical responses (Wang, 2026; Hu et al., 2026). Faithfulness studies perturb prompts, chains or parameters (Turpin et al., 2023; Lanham et al., 2023;

Chen et al., 2025; Tutek et al., 2025); we combine a known conflicting cue with internal readouts. Positional copying and output priors can also explain arithmetic errors (Liu, 2026; Prabhakar et al., 2024); Section 3.2 checks repetition of the digit preceding the answer. A companion code study (Anonymous, 2026) tests readouts before actual wrong value writes in Python traces and changes the demonstration format.

6 Conclusion

Writing and evaluating equations limits answers suggested by misleading names in almost every tested arithmetic contrast. Paired renaming measures this benefit directly; the written-value and probe controls guide further tests of how it arises.

Limitations

- **Task and models.** We test short arithmetic problems, English number-word names and models up to 12B parameters. Names such as q4 were tested at only one level. Real code, larger models, chat prompts and longer reasoning are untested. Worked examples affect accuracy; we require agreement across three sets.
- **What internal predictions mean.** Recovering a correct digit from internal states does not show that the model uses it to answer. Most tests supply correct calculations. On models’ own calculations, predictors recover digits reliably on ordinary-name problems for Llama-3.2-3B but not OLMo-2-1B, even after retraining; only two models are checked (Appendix F). Across nine models, better probe accuracy goes with fewer name errors. Removing OLMo reverses that pattern. Differences in arithmetic ability could explain the relationship. Two variables from one model do not count as independent models.
- **Why writing calculations helps.** Replacing internal states changes more than name information and can spoil correct answers. Calculations also repeat names and equations. Writing values alone lowers accuracy (Appendix B.4), so it cannot isolate why writing equations helps.

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Name	Problem (answer: 6)
Neutral	pen=1 + cup, cup=2 + 3; pen=?
Congruent	six=1 + cup, cup=2 + 3; six=?
Incongruent	four=1 + cup, cup=2 + 3; four=?

Table 2: Matched names preserve the arithmetic. The misleading digit, 4, occurs nowhere else.

Level	Problem (neutral names)	Steps	Held	Distr.
1	pen=1 + cup, cup=2; pen=?	1	1	0
2	pen=2 + 3, cup=1 + pen; cup=?	2	0	0
3	pen=1 + cup, cup=2 + 3; pen=?	2	1	0
4	pen=1 + cup, cup=2 + 3, box=4 + 5; pen=?	2	1	1
5	pen=1 + cup, cup=2 + box, box=1 + 2; pen=?	3	2	0

Table 3: The five levels of Kudo et al. (2026): additions or subtractions needed (*Steps*), variables unresolved when first read (*Held*), and unused equations (*Distr.*). At level 1 the intermediate value is stated rather than computed.

A Task and reproducibility

A problem contains single-digit variables, addition or subtraction, and a query. The five levels vary dependency order, depth and the presence of an unused equation (Table 3). Three same-level worked examples precede each test problem. The evaluated models are Gemma-3-12B-it, Gemma-3-4B, Gemma-3-4B-it, Llama-3.1-8B, Llama-3.1-8B-Instruct, Llama-3.2-3B, Llama-3.2-3B-Instruct, OLMo-2-1B-Instruct, OLMo-2-7B-Instruct, plus four derived checkpoints (Appendix B.3). Demonstrations use ordinary names for every word-name condition, with a reserved word pool disjoint from test names. Generation is greedy; only the test variable name changes between matched conditions.

We reimplement Kudo et al. (2026)’s task and probe recipe rather than using their code. The shared choices are single-digit outputs, same-level demonstrations, a 10-way linear residual-stream classifier, full-batch SGD at 10^{-3} for 10,000 epochs, and their span-by-layer-window patching design. For replication we train on letter names. The Llama-3.2-3B value-writing positions agree with their Table 3: equation 5 for the queried variable and 2 for the intermediate. Our additions are the conflicting names, matched base-lines, demonstration-set claim rule and equivalence bound.

Pre-chain probe maxima are 30.4 and 47.9, com-

pared with 17.8 and 33.2 in Kudo et al. (2026). The value-writing positions agree, while the accuracy difference may reflect training details.

A.1 Tokenizer checks

We test candidate names in five tokenizer contexts and retain words that form one token without merging with a neighbor. For Llama 3, all ten number words and 95 of 112 neutral candidates pass. Gemma and OLMo use the same lists; run-time checks confirm equal token lengths and positions within matched groups. Spaces around binary operators prevent merges such as 1-two. No instance in the reported runs is excluded by the layout check.

Data, compute and software. Each level has 2,000 matched sets, each rendered in every condition, and 10,000 separately generated problems for training probes; every SGD probe is trained with 3 seeds. Every run here is inference and the probes are linear; the three adapter checkpoints of Appendix B.3 were trained in a separate project, and that compute is not counted below. The recorded allocation time totals at most 185 GPU-hours on NVIDIA A30, H100 and H200 cards, including idle time and shared workers that also ran companion-code experiments. The models are from the Llama 3 (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024) families, with the reasoning-tuned checkpoint from Bercovich et al. (2025), each used under its release licence for research. Models run in PyTorch (Paszke et al., 2019) with Transformers (Wolf et al., 2020), and the probes are trained in PyTorch; the L-BFGS refit of Appendix C.1 uses scikit-learn (Pedregosa et al., 2011), and the mixed-effects check uses statsmodels (Seabold and Perktold, 2010).

B Full behavioral results and controls

B.1 Equivalence and accuracy floor

Following preregistration Amendment 3, equivalence requires a 90% cluster-bootstrap interval wholly inside the ± 2 -point margin (two one-sided 5% tests). We resample matched sets with their demonstration-seed replicates together, using 4,000 bootstrap draws. Other behavioral intervals remain at 95%. For CoT accuracy, 61 of 88 facilitation and interference contrasts meet the same equivalence bound. Of the 15 reliable accuracy contrasts, 5 are also equivalent within the margin

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
OLMo-2-1B-Instruct	CoT	47 [44,51]	-5.7*	+0.4	+7.0*	+3.2*
OLMo-2-1B-Instruct	direct	34 [30,38]	-0.0	+2.7*	+3.0*	-0.2

Table 4: The level-3 model below the preregistered 90% mean neutral-CoT accuracy floor, excluded from the main behavioral table. Columns and claim markers follow Table 1.

and 10 are not; a detectable effect can still be small. The written-value control uses the same matched pairs for its excess estimate and interval; the exception model’s estimate pools all three demonstration sets and clusters on matched set.

Contrast	Cells	Equivalent	Reliable
CoT name errors	44	43	3
CoT accuracy	88	61	15
Direct name errors	44	31	13

Table 5: Equivalence within the two-point margin uses 90% intervals; reliability uses 95% intervals excluding zero and the same sign across all three demonstration sets. Each accuracy cell is one facilitation or interference contrast.

B.2 Naming and written-value controls

Written values. We split generated chains by whether they state the target’s true value. An observation enters the excess estimate only when its neutral twin falls in the same split. The estimate and its 95% interval use those same paired observations. Pooled over eligible cells, the excess is +0.08 points in 132 cells where the value is stated and -0.10 in 27 where it is not. The exception persists in the stated-value split (main text), so this split does not explain its effect.

Identifier-style names. At level 3, four models receive names such as q4 instead of four, under three demonstration sets. Without CoT, queried-variable excess name errors range from -0.2 to +0.2 points; 0 of 4 cells are reliable. For number words on the same models, the range is -0.2 to +2.8, with 3 of 4 reliable. The numeric name-error effect is therefore stronger for number words in this comparison; congruent identifier names can still improve accuracy.

Word-class effects. With CoT at level 4, a number word on the queried variable costs Llama-3.2-3B 9.9 accuracy points whether the name agrees with the value or not. Those errors restate the

wrong equation. This is distinct from answering the digit the name denotes, which is why the name-suggested value result cannot be read as removal of every naming effect.

Unused variables. A number word on the level 4 distractor changes accuracy by at most 1.2 points. Its no-CoT excess name errors range from +2.1 to +2.3; 2 of 2 cells are reliable. With CoT, the range is +0.0 to +1.0, with 0 of 2 reliable.

Mixed-effects check. We compare the paired bootstrap with a mixed logit model, $\text{name_error} \sim \text{incongruent} + C(\text{seed})$, with a random intercept for matched set. Each twin’s outcome refers to the same name-suggested digit. Both procedures use the same fixed subsample of 600 sets per cell. Of 57 fitted cells, 50 agree on whether the interval excludes zero; 11 additional cells have no outcome variation. The mixed fit supports an effect in 5 cells where the bootstrap does not and withdraws support in 2 cells where it does (Gemma-3-12B-it, value-only chain, queried variable; Llama-3.1-Nemotron-Nano-8B, CoT, intermediate variable). Subsampling also removes support from 1 starred Table 1 cell under both procedures (Gemma-3-4B-it, no CoT, queried variable). The mixed model uses variational Bayes, so its interval is an approximate posterior interval. We retain the full-data bootstrap as the primary estimate on the reported percentage-point scale.

Positional-copy control. Across 4110 generated CoT name errors over models, levels and demonstration sets, the final number in the comma-delimited segment before the answer is the name-suggested value in 13% of cases and the target’s true value in 11%. A trailing name-suggested value is therefore absent from many errors. This check constrains the positional-copy explanation of Liu (2026), but does not identify what was copied in each case.

Model	Level	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
				helps	errors	helps	errors
Llama-3.1-8B	1	CoT	99 [98,100]	+0.0	+0.0	+0.3	+0.0
Llama-3.1-8B	1	direct	74 [38,99]	+16.1*	+11.2*	+8.8*	+1.3
Llama-3.2-3B	1	CoT	97 [95,100]	+1.1	+0.1	-2.4	+0.3
Llama-3.2-3B	1	direct	46 [21,93]	+0.6	+5.9*	-3.9*	+2.6*
Llama-3.1-8B	2	CoT	100 [100,100]	-0.1	+0.0	-0.0	+0.0
Llama-3.1-8B	2	direct	36 [33,37]	+11.8*	+0.7	+2.6	+2.2
Llama-3.2-3B	2	CoT	94 [90,98]	+3.7	+0.5*	+1.0*	-0.1
Llama-3.2-3B	2	direct	29 [27,30]	+10.8*	+2.5*	+2.6*	+2.2
Llama-3.1-8B	4	CoT	88 [63,100]	+1.0	-0.0	-2.3	+0.2
Llama-3.1-8B	4	direct	28 [26,31]	+13.7*	+5.8	+6.0*	+0.2
Llama-3.2-3B	4	CoT	64 [26,85]	-10.2*	+0.7	-2.6*	+0.4
Llama-3.2-3B	4	direct	20 [16,23]	-1.6	+1.9	-0.2	+1.2*
Llama-3.1-8B	5	CoT	65 [43,100]	+1.2	-0.3	+0.1	-0.1
Llama-3.1-8B	5	direct	20 [13,26]	+20.2*	+20.5*	+0.9	+0.4
Llama-3.2-3B	5	CoT	69 [48,97]	-1.4	+0.1	-2.5	-0.1
Llama-3.2-3B	5	direct	12 [11,12]	+4.7*	+3.5*	+1.1*	+0.5

Table 6: Additional task levels for the two Llama models. Columns and claim markers follow Table 1; level 1 states the intermediate value directly.

B.3 Derived checkpoints

We compare six checkpoints on the Llama-3.1-8B architecture: pretrained, instruction-tuned, single-task LoRA, multi-task LoRA, TIES-merged adapters and reasoning-tuned. Adapters are folded into the weights before evaluation. Shared tokenizers and identical matched problems permit paired comparisons (Table 7). A pretrained/instruct Gemma pair supplies a second family comparison.

Task accuracy complicates interpretation. The multi-task model’s reliable negative excess (-0.58) accompanies lower neutral accuracy (46.1%, versus 52.2%). The reasoning-tuned checkpoint has the lowest accuracy (39.4%). It also requires a system instruction to enable long-form reasoning; our raw-completion prompts leave that mode off. The result concerns the checkpoint under this prompt, not reasoning mode in deployment. Across these checkpoints, CoT excess name errors are at most 0.20 points in magnitude.

B.4 Values without equations

Our chain restates each equation before substituting into it, as Kudo et al.’s does. Their Tables 5–6 also report a format that writes only the sub-

results, cup=5, pen=6. We test this format at level 3 with three demonstration sets. Two predictions were registered for both Llama models: that the excess name errors at the intermediate’s value step would exceed the 2-point margin, and that the value would still be decodable there (probe accuracy at least 0.85), making any failure one of readout.

Both predictions fail as registered. Llama-3.1-8B writes the name-suggested value at the intermediate’s value step +0.15 points above its twin, inside the margin. Llama-3.2-3B is at +9.20, which passes our claim rule, but across the three demonstration sets that is +1.7 / +24.2 / +1.7, so one set carries most of it. More informative is what the probes say. At the step that writes the value, the true value is decodable at 0.81 and 0.59 for Llama-3.1-8B and at 0.61 and 0.79 for Llama-3.2-3B, against 1.00 and 0.99 for the same model under the full chain. Neutral accuracy falls with it, to 54% and 31%.

Removing the equations lowers both accuracy and true-value probe readouts. It therefore fails to isolate a readout error while preserving those measures.

Tuning	Model	Queried	Intermediate	Acc.
base	Llama-3.1-8B	+1.20*	-0.02	53.4
instruct	+ instruct	+0.15	+0.22	52.2
single-task finetune	+ task LoRA	+0.05	+0.23	52.2
multi-task finetune	+ multi-task LoRA	-0.58*	-0.08	46.1
weight merge	+ TIES merge	-0.10	+0.10	51.9
reasoning-tuned	Nemotron-Nano-8B	+1.35*	+0.33	39.4
base	Gemma-3-4B	+0.08	+0.02	60.2
instruct	+ instruct	+0.88*	+0.33	54.9

Table 7: Level 3, no CoT, name errors above the matched twin’s rate, mean over three demonstration sets. *Acc.* is neutral accuracy. * as in Table 1. The bottom block repeats the pretrained-to-instruct step in a second family, Gemma-3-4B.

C Probe and patching controls

Probes predict the true digit, the name-suggested digit and their probability margin. The name-identity control assigns a fixed random digit to each name; selectivity subtracts its accuracy from true-value accuracy. Layers in the supplied-calculation analysis are selected by neutral selectivity, averaged over three probe seeds. P1 is the name token, P2 the end of its defining equation, P3 the query, P4 the position before its value is written, and P5 the position before the final answer. P1 is a reference, and P3 can reveal name identity rather than a computed value. Figure 3 gives the token and layer sweep supporting the main readout analysis.

Patches replace residual-stream activations at matched name-token positions, at individual layers or across all layers. The neutral-twin source tests removal of the name error; a different ordinary name tests damage to correct answers; an alternative misleading name tests specificity. The span sweep instead replaces an equation, the query or the value-writing segment in four-layer windows. Figure 4 supports the early-layer localization. Table 8 gives the answer-only control rates and denominators for both Llama models. OLMo’s supplied-calculation controls are reported in Table 10. We also reverse the main contrast, transplanting misleading-name states into the ordinary-name twin. OLMo’s largest injection rate across layers and both variables is 2.7% at layer 5; word-control damage there is 0.4%. The injection itself damages 7.0% of correct answers. This selected maximum is descriptive.

Instance-level association. Within each model, level, regime, variable and position, we regress name errors on the probe margin, $\log p_{\text{true}} - \log p_{\text{name}}$, where p_{name} is the probe probability

of the name-suggested value, with standard errors clustered on matched set. The layer is selected by neutral accuracy. Of 63 fits, 29 have $p < .05$: 24 coefficients are negative and 5 positive. The direction is not consistent, and 4 positive results come from the model supplying most fits. This leaves the instance-level association unresolved. The model-level pattern is also sensitive to panel composition (Appendix D).

C.1 Probe recipe and patching scope

We check two implementation choices inherited from Kudo et al. (2026): the probe optimizer and the name occurrences included in a patch.

Recipe. We refit the level-3 probes for Llama-3.2-3B with L-BFGS, and with SGD on standardized inputs, against the reported recipe.

All three recipes give the same qualitative result: the value is decodable where the chain writes it (0.96–1.00) and not at the end of its defining equation (0.18–0.50). The query position moves, from 0.271 under the reported recipe to 0.393 under standardization, within a range of 0.18–0.39 over recipes and variables. We already report that position as the presence of the name rather than a decoded value, because its control task scores 0.97, so this variation does not change the interpretation of that position.

Scope. Patching every occurrence of the name includes the copy the model writes in its own output. In the direct regime, restricting the patch to the name’s occurrences in the prompt leaves the intermediate variable essentially unchanged, since its name never appears in the answer: removal is 43% at layers 0–3 under both scopes. For the queried variable the best windows remove similar shares (52% at layers 4–7 prompt-only, against 53% at layers 0–3), the prompt-only patch re-

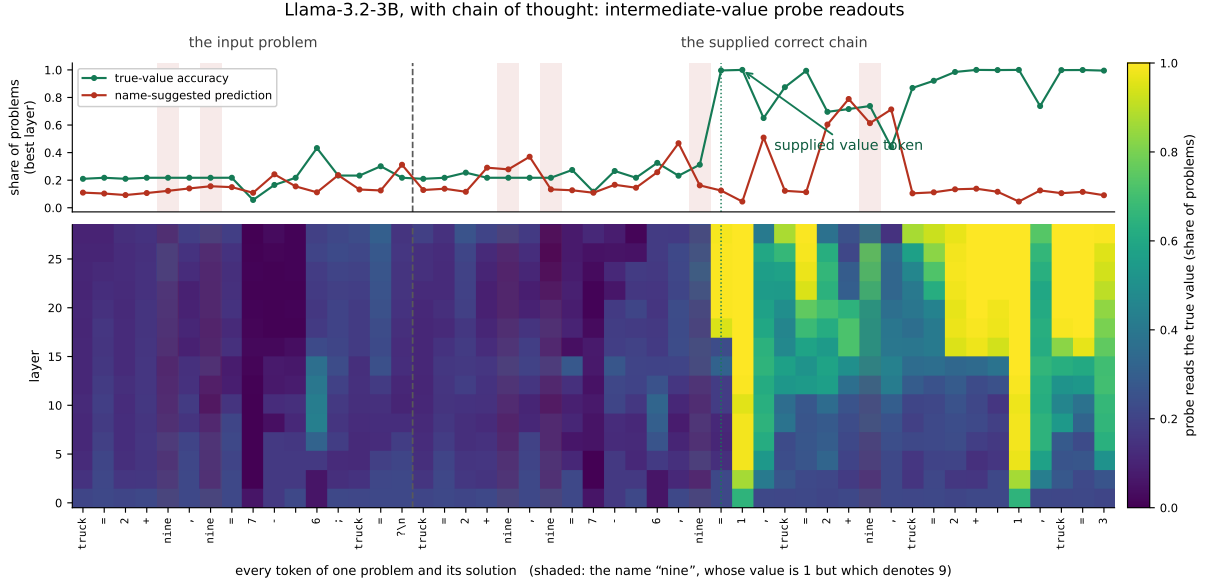


Figure 3: Llama-3.2-3B, intermediate variable under supplied correct calculations. Top: best-layer correct-digit accuracy and name-suggested predictions; bottom: correct-digit accuracy by layer. Rates pool problems; tokens illustrate one example.

Model	Variable	Pairs	Name errors	Correct	Neutral	Alternative	Word damage
Llama-3.2-3B	queried	2000	172	272	51.7	34.9	22.4
Llama-3.1-8B	intermediate	2000	72	838	30.6	30.6	8.5

Table 8: All-layer answer-only patches, demonstration set 7. Neutral and alternative: percentage of baseline name errors removed. Word damage: percentage of correct ordinary-name answers changed; *Correct* gives that denominator.

moves less in the later windows, and the two disagree most in the first window, where the prompt-only patch removes 3% and damages 76% of correct answers. That is mechanical: the queried variable’s name also sits at the position we read, so patching its prompt copies alone leaves the model reading a name whose early representation no longer matches. The localization we report does not depend on the choice.

D Task competence and value readouts

Table 11 compares both variables for 9 models. Layers are selected by neutral selectivity, averaging three probe seeds. Task accuracy and paired excess name errors pool demonstration sets 7, 11 and 13; probes use set 7. The within-model difference holds overall task accuracy fixed but does not isolate a causal effect of the readout. We average the two variable readouts within each model before computing descriptive correlations. Table 12 repeats them after omitting each model. Roles are not independent observations. With this small, het-

erogeneous panel, correlations cannot distinguish overall competence from the role of decodable information.

E Gold-trained transfer to generated chains

We read the state before the first numeric assignment to the target in each saved generated chain. Expression operands are not counted as completed value writes; tokens that include any part of the supplied value are excluded. A fixed sample of 200 matched sets is selected by a hash independent of outcomes, shared across models, variables and demonstration sets. All additional chains with name errors and their neutral twins are evaluated separately. Neutral gold-chain probes use 10,000 separate training problems, three probe seeds and the original neutral-selectivity layer. Table 13 tests transfer to held-out neutral generated chains. A high-accuracy interpretation requires at least 90% neutral calibration in that demonstration set. Cases below that threshold yield descriptive readouts

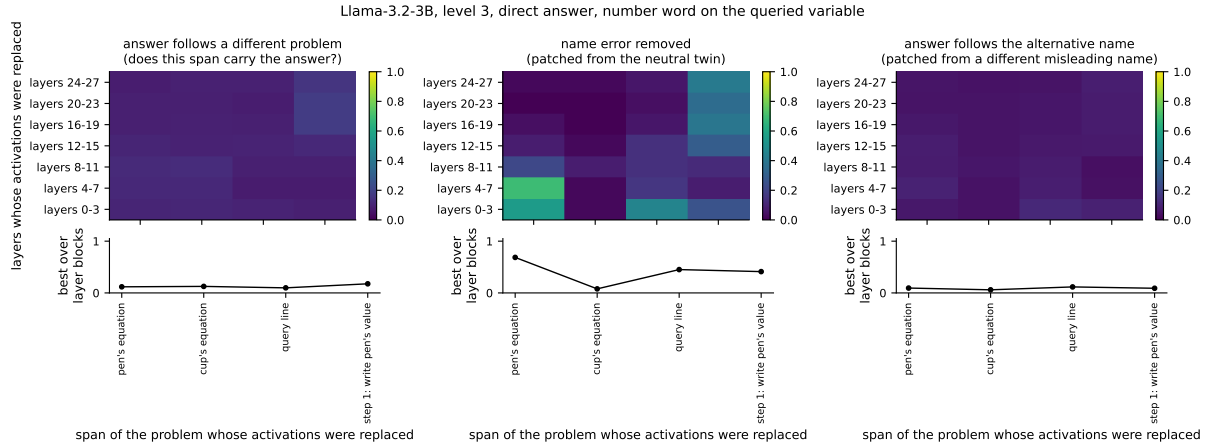


Figure 4: Llama-3.2-3B, queried variable, answer-only. Equation spans are replaced in four-layer windows. Sources: a different problem, the ordinary-name twin, or an alternative misleading-name twin. Bottom: maximum over layer windows.

Position	Variable	SGD (reported)	L-BFGS	standardized
defining equation	queried	0.184	0.190	0.183
defining equation	intermediate	0.496	0.422	0.420
query	queried	0.183	0.316	0.280
query	intermediate	0.271	0.366	0.393
value step	queried	1.000	1.000	1.000
value step	intermediate	0.982	0.962	0.972
answer	queried	1.000	1.000	1.000

Table 9: Best-layer true-value accuracy on neutral problems under three probe recipes (proportions from 0 to 1).

only. The augmentation is selected by observed errors and cannot estimate their population rate. Appendix F separately trains probes on generated neutral chains and tests the same held-out prefixes. Llama-3.2-3B meets the neutral transfer threshold in 6 of 6 cells, with calibration accuracy from 97.2 to 100.0%. OLMo-2-1B meets the neutral transfer threshold in 0 of 6 cells, with calibration accuracy from 19.4 to 62.2%. Name-error cases in the competent model are few; error-conditioned read-outs cannot establish a general mechanism from that subset. Failed neutral calibration prevents a strong generated-chain interpretation for the exception model.

F Training probes on generated neutral chains

Gold-trained probe transfer (Appendix E) may fail because the states along generated chains differ from those along the supplied correct chain. We therefore train on freely generated neutral chains from 2,000 source problems and select layers on 400 separate validation problems. Canonical equations partition the source pool, keeping renamed versions of a computation together. The training

and validation pools contain 630 and 155 computations; both exclude all test and demonstration computations. These counts distinguish independent computations from repeated name renderings. Generation is greedy with demonstration set 7 and a 128-token limit. The probe sees the hidden state before the target variable’s first numeric assignment, with every token boundary checked to exclude the value digit. Missing or unsafe boundaries are excluded and counted. We retain all usable neutral chains regardless of whether the model writes the correct value, using the executed true value as the training label. Three probe seeds use the original full-batch SGD recipe (learning rate 0.001, 10,000 epochs). States are stored in float32 and nonfinite states or fitted weights are rejected. Layers are searched at stride two, including the final layer, and chosen only by mean validation accuracy with the lowest-layer tie break. The name-identity control is trained separately and does not select layers. Table 14 reports the usable source counts and selected layers. Evaluation uses exactly the earlier fixed 200-set sample and separately reported additional name-error cases under demonstration sets 7, 11 and

Target	Read	Errors	Neutral removed	Word dmg.	Alt. removed	Alt. followed
v1	value	0	–	1.6	–	0.0
v1	answer	0	–	1.6	–	0.0
v2	value	28	92.9	7.3	82.1	4.0
v2	answer	3	100.0	0.8	100.0	0.6

Table 10: OLMo-2-1B-Instruct, supplied correct calculations, 500 matched pairs per target. All-layer rates (%): neutral/alternative name-error removal, word-control damage to correct predictions, and predictions following the alternative digit. Roles v1/v2: queried/intermediate; –: undefined.

Model	Role	Layer	Task	Neutral	Name ID	Misleading	Excess
Gemma-3-12B-I	v1	40	100.0	100.0	13.1	100.0	0.0
Gemma-3-12B-I	v2	42	100.0	100.0	11.9	100.0	0.0
Gemma-3-4B	v1	25	100.0	100.0	10.0	100.0	0.0
Gemma-3-4B	v2	28	100.0	99.8	12.4	100.0	0.0
Gemma-3-4B-I	v1	29	100.0	100.0	7.4	99.9	0.0
Gemma-3-4B-I	v2	28	100.0	99.8	10.0	99.9	0.0
Llama-3.1-8B	v1	20	95.2	100.0	20.5	100.0	-0.1
Llama-3.1-8B	v2	22	95.2	99.4	24.9	99.2	-0.0
Llama-3.1-8B-I	v1	20	99.7	100.0	21.0	100.0	0.0
Llama-3.1-8B-I	v2	20	99.7	100.0	25.7	100.0	0.0
Llama-3.2-3B	v1	19	98.3	99.9	22.7	100.0	0.1
Llama-3.2-3B	v2	21	98.3	97.6	31.5	99.6	-0.2
Llama-3.2-3B-I	v1	18	100.0	99.9	26.3	100.0	0.0
Llama-3.2-3B-I	v2	19	100.0	99.9	29.5	99.9	0.0
OLMo-2-1B	v1	14	47.2	99.7	48.9	100.0	0.4
OLMo-2-1B	v2	0	47.2	21.8	15.0	21.8	3.2
OLMo-2-7B	v1	22	95.3	100.0	37.7	100.0	-0.1
OLMo-2-7B	v2	26	95.3	99.4	47.5	97.6	-0.0

Table 11: Task and true-digit probe accuracies (%), name-identity control accuracy (%), and excess name errors (points). Roles v1/v2 are queried/intermediate. Selectivity is neutral minus name-ID accuracy. Below-floor models remain descriptive.

13. Table 15 gives held-out neutral calibration and boundary coverage. Intervals use 4,000 canonical-computation bootstrap draws, keeping repeated names together. The 90% calibration threshold applies to the point estimate, not the lower interval endpoint. Llama passes all six calibration cells; OLMo passes none. Table 16 retains error readouts with their denominators. OLMo’s readouts remain descriptive. Llama has only one name-error case in the primary sample and thirteen additional observed writes; these rare, selected cases cannot establish a general mechanism.

G Paired change from answer-only to written calculations

We compare prompting conditions on exactly the same matched sets and demonstration seeds used in the 44 name-error equivalence cells. Each observation subtracts the ordinary-name rate from the misleading-name rate, then subtracts the answer-only contrast from the CoT contrast. Nega-

tive changes indicate fewer added name-suggested answers. We bootstrap matched sets, keeping all three demonstration repeats together (4,000 draws). Reliable reductions require a negative change in every demonstration set and a 95% interval below zero. The panel is descriptive across related checkpoints; its cells are not independent model replications. Of the 44 paired changes, 10 meet that rule; 19 have intervals below zero. A sensitivity bootstrap that also clusters repeated arithmetic computations yields 10 reliable reductions. Intervals are pointwise; the count is not a multiplicity-corrected family-wide test. Before writing calculations, 31 cells already meet the two-point equivalence bound, compared with 43 afterward. Thus the equivalence count is not a count of significant improvements. Accuracy gains accompany many reductions and do not identify an effect independent of arithmetic competence.

Omitted model	Task-excess	Probe-excess	Task-probe
All models	-0.99	-1.00	+1.00
Gemma-3-12B-I	-0.99	-1.00	+1.00
Gemma-3-4B	-0.99	-1.00	+1.00
Gemma-3-4B-I	-0.99	-1.00	+1.00
Llama-3.1-8B	-0.99	-1.00	+1.00
Llama-3.1-8B-I	-0.99	-1.00	+1.00
Llama-3.2-3B	-0.99	-1.00	+1.00
Llama-3.2-3B-I	-0.99	-1.00	+1.00
OLMo-2-1B	+0.98	+0.78	+0.85
OLMo-2-7B	-0.99	-1.00	+1.00

Table 12: Descriptive Pearson correlations of model means; Spearman values and within-model variable differences are in the saved summary. No significance or causal claim is based on these coefficients.

Model	Set	Role	Neutral coverage	Calibration (%)	Sample errors	Extra errors
OLMo-2-1B	7	v1	194/200	62.2 [55.2, 68.7]	8	69
OLMo-2-1B	7	v2	186/200	19.4 [14.0, 25.3]	11	82
OLMo-2-1B	11	v1	187/200	55.3 [48.3, 62.6]	7	59
OLMo-2-1B	11	v2	129/200	21.7 [14.7, 28.7]	2	21
OLMo-2-1B	13	v1	193/200	52.7 [45.6, 59.8]	17	159
OLMo-2-1B	13	v2	124/200	24.2 [16.9, 32.3]	1	30
Llama-3.2-3B	7	v1	200/200	98.0 [96.0, 99.5]	1	8
Llama-3.2-3B	7	v2	200/200	97.2 [94.8, 99.0]	0	0
Llama-3.2-3B	11	v1	199/200	100.0 [100.0, 100.0]	0	0
Llama-3.2-3B	11	v2	200/200	100.0 [100.0, 100.0]	0	0
Llama-3.2-3B	13	v1	199/200	99.5 [98.5, 100.0]	0	5
Llama-3.2-3B	13	v2	200/200	99.5 [98.5, 100.0]	0	0

Table 13: Gold-trained probe accuracy on neutral generated chains [95% interval], and eligible name-error counts. Extra cases exclude the fixed sample. Roles v1/v2 are queried/intermediate. Calibration is required before interpreting error readouts.

Model	Role	Train n	Validation n	Layer	Validation (%)	Name ID (%)
OLMo-2-1B-I	v1	1924	390	14	47.9	41.6
OLMo-2-1B-I	v2	1877	376	16	87.8	86.6
Llama-3.2-3B	v1	1994	398	22	97.5	26.8
Llama-3.2-3B	v2	2000	400	20	99.2	28.6

Table 14: Usable generated-boundary counts and validation-selected layers. Roles v1/v2 are queried/intermediate. Source cohorts are computation-disjoint; validation scores are not test estimates.

Model	Role	Set	Neutral coverage	Calibration (%)	Name ID (%)	Pass
OLMo-2-1B-I	v1	7	194/200	61.3 [54.1, 68.6]	37.3 [31.2, 43.4]	no
OLMo-2-1B-I	v1	11	187/200	52.8 [45.1, 60.3]	32.4 [26.6, 38.5]	no
OLMo-2-1B-I	v1	13	193/200	50.3 [43.0, 57.4]	35.2 [29.1, 41.6]	no
OLMo-2-1B-I	v2	7	186/200	78.0 [71.5, 83.9]	81.9 [76.9, 86.7]	no
OLMo-2-1B-I	v2	11	129/200	80.1 [72.8, 86.8]	56.6 [49.6, 63.3]	no
OLMo-2-1B-I	v2	13	124/200	71.8 [64.2, 78.4]	46.8 [37.8, 55.6]	no
Llama-3.2-3B	v1	7	200/200	98.5 [96.7, 100.0]	18.5 [13.8, 23.7]	yes
Llama-3.2-3B	v1	11	199/200	100.0 [100.0, 100.0]	13.9 [9.5, 18.5]	yes
Llama-3.2-3B	v1	13	199/200	97.2 [94.2, 99.5]	16.9 [12.3, 21.8]	yes
Llama-3.2-3B	v2	7	200/200	96.3 [93.6, 98.6]	28.8 [22.7, 35.0]	yes
Llama-3.2-3B	v2	11	200/200	100.0 [100.0, 100.0]	17.2 [12.4, 22.1]	yes
Llama-3.2-3B	v2	13	200/200	97.7 [95.5, 99.3]	17.8 [13.3, 22.8]	yes

Table 15: Held-out primary neutral generated chains. Coverage counts valid pre-value boundaries over all primary neutral rows. Accuracy intervals are 95% canonical-computation bootstrap intervals; pass means the calibration point estimate meets 90%.

Model	Role	Set	Sample n	Sample accuracy	Extra n	Extra accuracy
OLMo-2-1B-I	v1	7	8	4.2 [0.0, 12.5]	69	2.9 [0.0, 7.0]
OLMo-2-1B-I	v1	11	7	19.0 [0.0, 38.1]	59	10.7 [4.2, 18.8]
OLMo-2-1B-I	v1	13	17	21.6 [7.8, 39.2]	159	13.6 [8.6, 19.4]
OLMo-2-1B-I	v2	7	11	33.3 [15.2, 51.5]	82	18.7 [11.4, 27.2]
OLMo-2-1B-I	v2	11	2	0.0 [0.0, 0.0]	21	17.5 [3.3, 31.8]
OLMo-2-1B-I	v2	13	1	0.0 [0.0, 0.0]	30	25.6 [13.7, 39.1]
Llama-3.2-3B	v1	7	1	0.0 [0.0, 0.0]	8	12.5 [0.0, 42.9]
Llama-3.2-3B	v1	11	0	–	0	–
Llama-3.2-3B	v1	13	0	–	5	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	0	–	0	–
Llama-3.2-3B	v2	11	0	–	0	–
Llama-3.2-3B	v2	13	0	–	0	–

Table 16: True-digit accuracy (%) [95% interval] on actual name errors, with the fixed sample and additional error-selected cases separated. Extra cases exclude the primary sample. Dashes denote empty strata. OLMo fails neutral calibration, so its readouts are descriptive.

Model	Level/role	Answer-only	CoT	Paired change [95% interval]	Accuracy gain
Llama-3.2-3B	L1/v1	+5.90	+0.12	-5.78 [-6.55, -5.02]*	+50.7
Llama-3.2-3B	L1/v2	+2.62	+0.30	-2.32 [-2.87, -1.77]*	+50.7
Llama-3.1-8B	L1/v1	+11.20	+0.00	-11.20 [-12.05, -10.37]*	+25.6
Llama-3.1-8B	L1/v2	+1.33	+0.02	-1.32 [-1.68, -0.97]	+25.6
Llama-3.2-3B	L2/v1	+2.50	+0.53	-1.97 [-2.65, -1.28]*	+65.1
Llama-3.2-3B	L2/v2	+2.20	-0.05	-2.25 [-2.92, -1.63]	+65.1
Llama-3.1-8B	L2/v1	+0.73	+0.02	-0.72 [-1.25, -0.20]	+64.4
Llama-3.1-8B	L2/v2	+2.20	+0.00	-2.20 [-2.73, -1.63]	+64.4
Llama-3.2-3B	L3/v1	+2.78	+0.13	-2.65 [-3.30, -2.00]*	+76.1
Llama-3.2-3B	L3/v2	+1.78	-0.17	-1.95 [-2.42, -1.48]	+76.1
Llama-3.2-3B-I	L3/v1	+0.17	+0.00	-0.17 [-0.62, +0.28]	+56.0
Llama-3.2-3B-I	L3/v2	+0.47	+0.00	-0.47 [-1.05, +0.12]	+56.0
Llama-3.1-8B	L3/v1	+1.20	-0.13	-1.33 [-1.82, -0.85]*	+41.8
Llama-3.1-8B	L3/v2	-0.02	-0.02	+0.00 [-0.47, +0.47]	+41.8
Llama-3.1-8B-I	L3/v1	+0.15	+0.00	-0.15 [-0.60, +0.28]	+47.5
Llama-3.1-8B-I	L3/v2	+0.22	+0.02	-0.20 [-0.68, +0.28]	+47.5
Gemma-3-4B-I	L3/v1	+0.88	+0.00	-0.88 [-1.38, -0.38]*	+45.1
Gemma-3-4B-I	L3/v2	+0.33	+0.02	-0.32 [-0.80, +0.15]	+45.1
Gemma-3-12B-I	L3/v1	-0.23	+0.00	+0.23 [-0.08, +0.55]	+22.4
Gemma-3-12B-I	L3/v2	+0.40	+0.00	-0.40 [-0.83, +0.02]	+22.4
OLMo-2-1B-I	L3/v1	+2.73	+0.45	-2.28 [-3.28, -1.30]*	+13.5
OLMo-2-1B-I	L3/v2	-0.20	+3.17	+3.37 [+2.37, +4.37]	+13.5

Table 17: Excess name errors and paired CoT-minus-answer-only changes (percentage points). Accuracy gain is the ordinary-name accuracy change. Roles v1/v2/v3 are the first/second/third variables in the problem; the queried role is v2 at level 2 and v1 otherwise. *Reliable reduction across all demonstration sets.

Model	Level/role	Answer-only	CoT	Paired change [95% interval]	Accuracy gain
OLMo-2-7B-I	L3/v1	-0.23	-0.08	+0.15 [-0.33, +0.60]	+41.0
OLMo-2-7B-I	L3/v2	-0.42	-0.03	+0.38 [-0.13, +0.87]	+41.0
Llama-8B single-task	L3/v1	+0.05	+0.00	-0.05 [-0.50, +0.42]	+47.6
Llama-8B single-task	L3/v2	+0.23	-0.02	-0.25 [-0.75, +0.23]	+47.6
Llama-8B multi-task	L3/v1	-0.58	+0.00	+0.58 [+0.23, +0.95]	+53.7
Llama-8B multi-task	L3/v2	-0.08	+0.00	+0.08 [-0.38, +0.53]	+53.7
Llama-8B merged	L3/v1	-0.10	+0.00	+0.10 [-0.30, +0.50]	+47.8
Llama-8B merged	L3/v2	+0.10	+0.00	-0.10 [-0.58, +0.38]	+47.8
Nemotron-Nano-8B	L3/v1	+1.35	+0.08	-1.27 [-1.78, -0.77]	+58.8
Nemotron-Nano-8B	L3/v2	+0.33	+0.20	-0.13 [-0.63, +0.38]	+58.8
Gemma-3-4B	L3/v1	+0.08	+0.00	-0.08 [-0.53, +0.35]	+39.8
Gemma-3-4B	L3/v2	+0.02	+0.00	-0.02 [-0.52, +0.48]	+39.8
Llama-3.2-3B	L4/v1	+1.93	+0.68	-1.25 [-1.78, -0.70]	+44.4
Llama-3.2-3B	L4/v2	+1.17	+0.37	-0.80 [-1.32, -0.27]	+44.4
Llama-3.1-8B	L4/v1	+5.83	-0.03	-5.87 [-6.62, -5.17]	+59.5
Llama-3.1-8B	L4/v2	+0.23	+0.20	-0.03 [-0.45, +0.37]	+59.5
Llama-3.2-3B	L5/v1	+3.45	+0.08	-3.37 [-4.10, -2.67]*	+57.5
Llama-3.2-3B	L5/v2	+0.52	-0.07	-0.58 [-1.17, +0.03]	+57.5
Llama-3.2-3B	L5/v3	-0.35	-0.08	+0.27 [-0.35, +0.88]	+57.5
Llama-3.1-8B	L5/v1	+20.50	-0.30	-20.80 [-21.85, -19.75]*	+45.1
Llama-3.1-8B	L5/v2	+0.45	-0.15	-0.60 [-1.28, +0.05]	+45.1
Llama-3.1-8B	L5/v3	+0.33	+0.08	-0.25 [-0.92, +0.42]	+45.1

Table 18: Excess name errors and paired CoT-minus-answer-only changes (percentage points). Accuracy gain is the ordinary-name accuracy change. Roles v1/v2/v3 are the first/second/third variables in the problem; the queried role is v2 at level 2 and v1 otherwise. *Reliable reduction across all demonstration sets.